VICTORY 2026

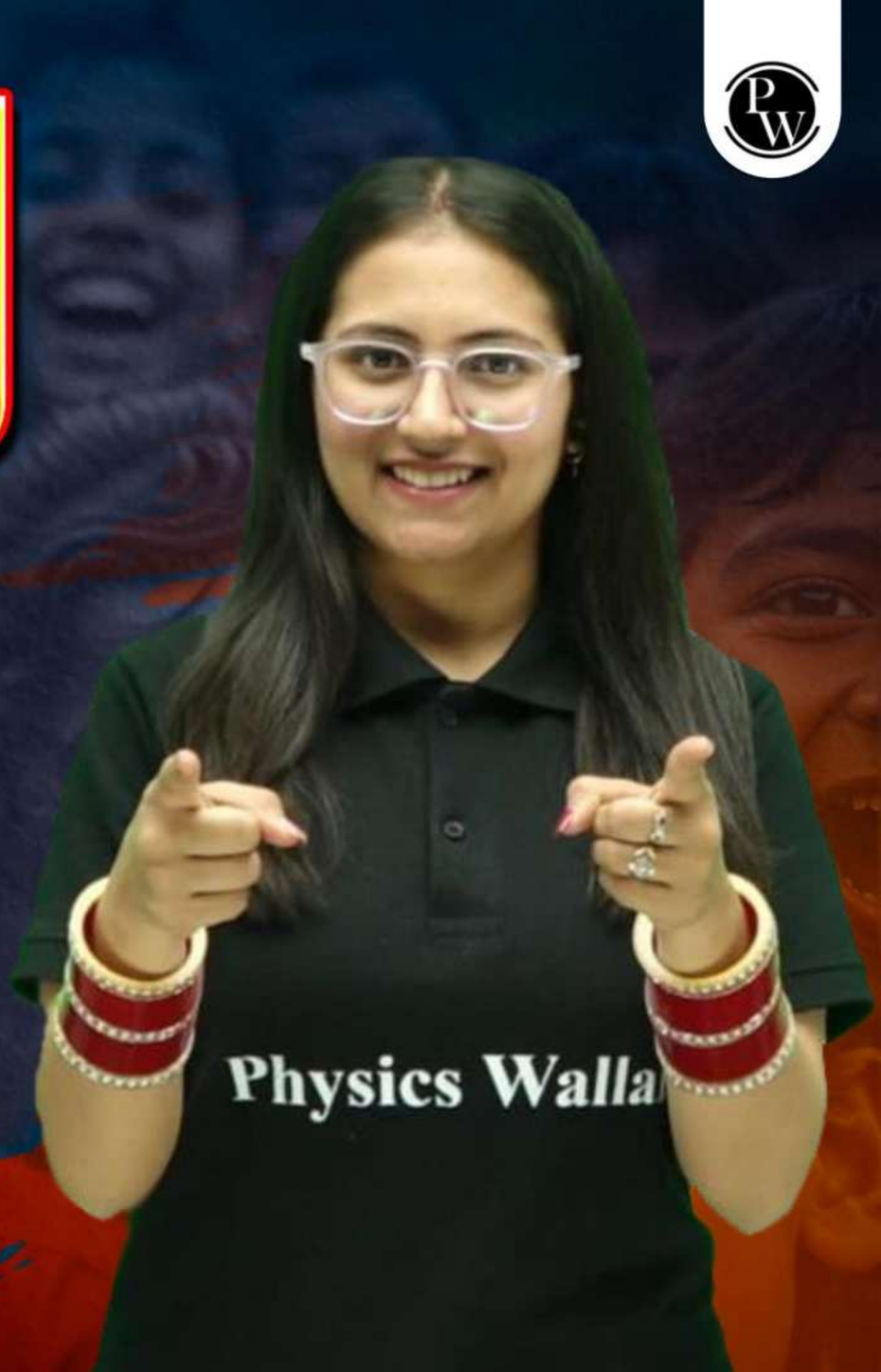
CHEMISTRY

Study of Compounds :

Nitric Acid (HNO_3)

Lect.-04

By- Bharti Ma'am



Topics to be covered

A

B

C

D

`Nitric acid : one-shot'

Nitric Acid 'HNO_3'

Chemical formula: HNO_3

(Strong Mineral acid)

HNO_2 (Nitrous Acid)

HNO_2 is aka **Aqua fortis** 'Strong water'

gold and silver dissolve in HNO_3



Occurance

1. Occurance

In free state, nitric acid does not occur in nature in significant quantities. However, it is found in rain through natural processes like lightning and microbial activities.

At high temperature after lightening it react with oxygen and form nitric oxide.

$$N_2 + O_2 \xrightarrow[\text{(lighening)}]{3000^{\circ}C} 2NO$$

2. Oxidation of nitric oxide:

Nitric oxide (NO) reacts with atmospheric oxygen to form nitrogen dioxide (NO_2).

$$2NO + O_2 \rightarrow 2NO_2$$

3. Dissolution in water:

Nitrogen dioxide (NO_2) dissolves in atmospheric water vapour to form nitric acid (HNO_3).

Fixation atmospheric nitrogen:

It is the process of conversion of molecular nitrogen (N_2) from the earth's atmosphere into useful nitrogen compound in the soil.

Method of Preparation of Nitric Acid (Laboratory Method)

Imp

Reactants Used:

- Potassium nitrate (KNO_3) or sodium nitrate ($NaNO_3$) and conc. H_2SO_4.

→ KNO_3 OR $NaNO_3$

→ Conc. H_2SO_4

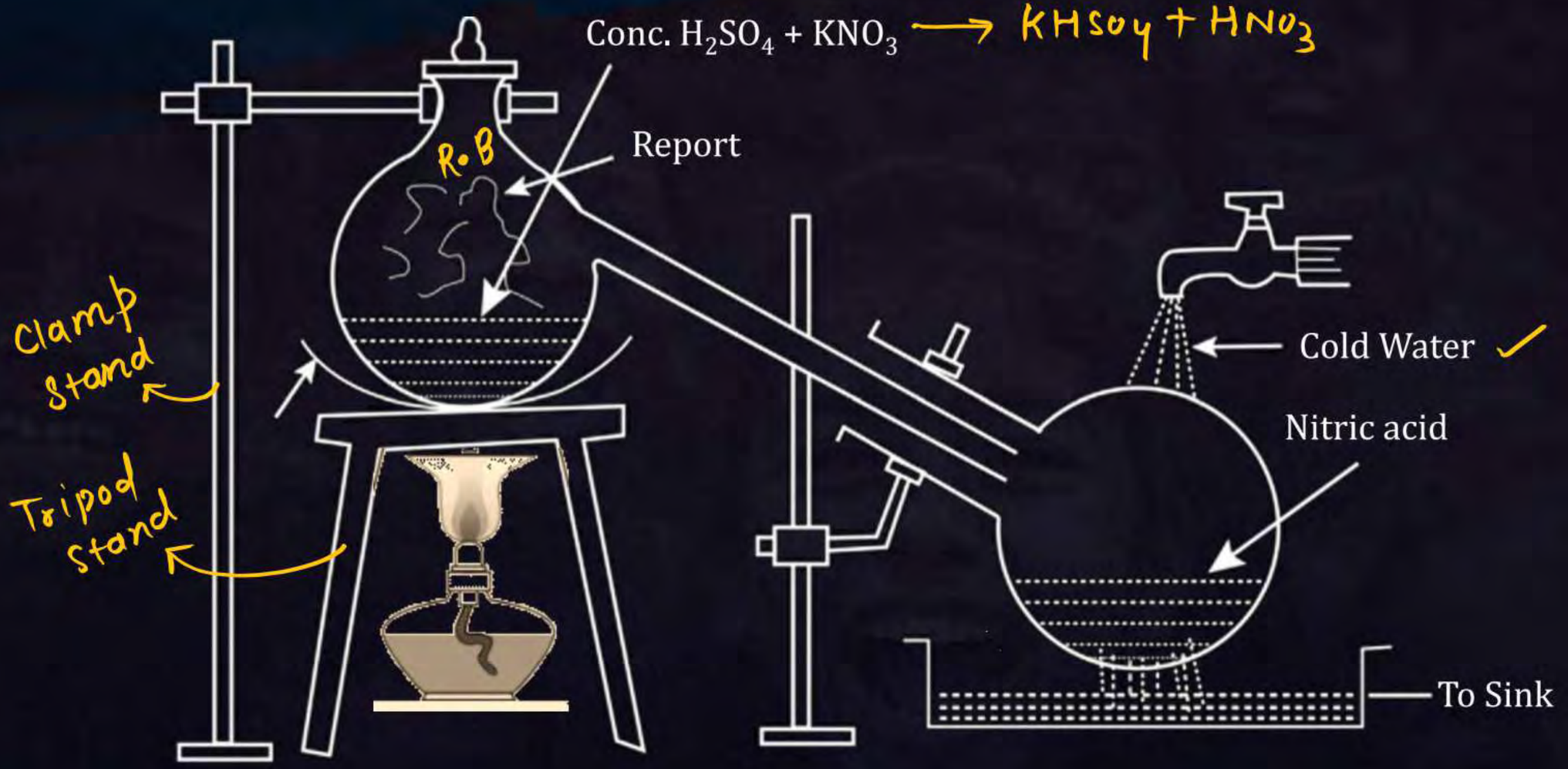
(a) $KNO_3 + H_2SO_4 \xrightarrow[\text{less than } 200^\circ C]{< 200^\circ C} KHSO_4 + HNO_3$ (Nitric acid)

(b) $NaNO_3 + H_2SO_4 \xrightarrow[\text{less than } 200^\circ C]{< 200^\circ C} NaHSO_4 + HNO_3$ (Nitric acid)

Method of Preparation of Nitric Acid (Laboratory Method)

Method:

- Take equal part of potassium nitrate (nitre) or sodium nitrate (chile salt petre) and concentrated sulphuric acid in a glass retort.
- Heat the mixture gently to 180°C-200°C. * Temp :- Not More than 200°C
- On heating the vapour of nitric acid are evolved (as sulphuric acid is a non-volatile acid does not produce vapour).
- The vapour of nitric acid enters in a a receiver and is kept cool in a stream of cold water



Laboratory preparation of nitric acid

Note:

Pure nitric acid is colourless liquid but in the laboratory methods, the nitric acid obtained exhibit light yellow or brown colour. This yellow or a brown colour arises due to dissolution of nitrogen dioxide gas. This gas is generated during the thermal decomposition of a fraction of nitric acid.

$$4HNO_3 \xrightarrow[\text{Decomposition}]{\text{Thermal}} \underset{\text{(Reddish brown fumes)}}{4NO_2} + 2H_2O + O_2$$

Imp

Imp



Removal of yellow colour from acid

- **The collected nitric acid is yellow (or brown) colour. This yellow (or brown) colour can be removed.**

i) By addition of excess of water: Addition of excess water causes dissolution of nitrogen dioxide gas which is soluble in water and thus yellow colour of acid is removed.

$$4NO_2 + O_2 + 2H_2O \rightarrow 4HNO_3$$



Removal of yellow colour from acid

- **The collected nitric acid is yellow (or brown) colour. This yellow (or brown) colour can be removed.**

ii) By passing carbon dioxide gas: Carbon dioxide (CO_2) drives out the reddish brown nitrogen dioxide gas from the warm acid and changes to colourless.

Imp

Removal of yellow colour from acid

- **The collected nitric acid is yellow (or brown) colour. This yellow (or brown) colour can be removed.**

iii) By passing air: On passing air, nitrogen dioxide oxidises and converted into nitric acid.

$$4NO_2 + 2H_2O + O_2 \rightarrow 4HNO_3$$

Oxidation of $NO_2 \longrightarrow HNO_3$

iski vjha se yellow color

Precautions:

1. Conc. HCl is volatile acid. Hence, in preparation of nitric acid it is not used in place of conc. H_2SO_4 because it may volatilise along with HCl vapours.

2. Use glassware and equipment that are resistant to the carrosive nature of nitric acid because vapours of nitric acid attack rubber and cork.

Precautions:

3. Monitor and control the reaction temperature as the formation of nitric acid can be exothermic. Therefore temperature of reaction should not be exceed 200°C because at higher temperature. Na_2SO_4

   i) A hard crust of sodium sulphate is formed which sticks to the wall of the retort and is difficult to remove.

   ii) The glass apparatus may break and fuel is wasted.

V. Imp.

Manufacture of Nitric Acid (By Ostwald's Process)

V. Imp

1. Ammonia oxidation:

A mixture of dry air (free from CO, and dust particles) and dry ammonia (NH_2) in the ratio of 10:1 (by volume) is first compressed and then passed into catalyst chamber. Here ammonia (NH_3) is oxidised using air in the presence of a platinum or rhodium catalyst at high temperature (i.e., about 800-900°C).

The reaction is as follows:

$$4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O + \text{Heat}$$

Exothermic Process!

$NH_3 \xrightarrow{\text{oxidn}} NO$ → Nitric oxide

10 : 1 by Volm

dry Air, $NH_3(g)$ → first compressed → Catalytic Chamber (Pt)

800°c - 900°c



Wilhelm Ostwald



Manufacture of Nitric Acid (By Ostwald's Process)

2. **Cooling of Gases:**

After the catalyst chamber the gases are passed through the cooling chamber where cooling pipes cool the gases at around 50°C temperature. Lower temperature reduces the decomposition of nitrogen dioxide and oxidises nitric oxide to nitrogen dioxide easily

Manufacture of Nitric Acid (By Ostwald's Process)

3. Formation of nitrogen dioxide:

Nitrogen dioxide formation occurs in oxidation chamber where nitrogen monoxide produced in first step reacts further with oxygen to form nitrogen dioxide (NO_2) at 50°C temperature.

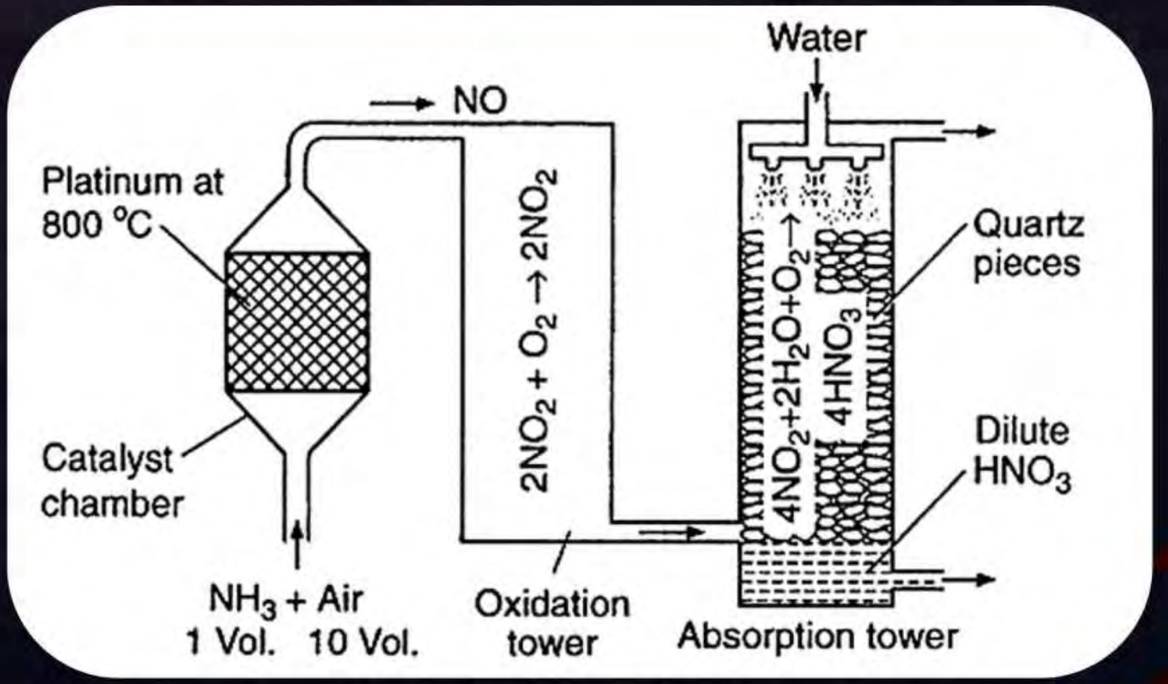
$$2NO + O_2 \xrightarrow{50^\circ C} 2NO_2$$

Manufacture of Nitric Acid (By Ostwald's Process)

4. Absorption in water:

The absorption of nitrogen dioxide in water takes place in absorption tower. Absorption tower is made up of stainless steel packed with acid resistant quartz.

$$4NO_2 + 2H_2O + O_2 \rightarrow 4HNO_3$$



Manufacture of nitric acid

Note: Imp

dry Air 10:1 why ?

- In manufacturing of nitric acid, larger quantity of air is needed because it is required all the three chambers.

- Quartz is acid resistant and help in dissolving nitrogen dioxide gas uniformly in water.

Imp

$NO_2 + H_2O + O_2 \longrightarrow$ ✓

NH_3 (1 vol) Air (10 vol)

800°C to 900°C

Catalyst chamber
(oxidation of Ammonia)

$$4NH_3 + 5O_2 \xrightarrow[800^\circ C]{Pt} 4NO + 6H_2O + Heat$$

$6H_2O$

↓

Heat Exchanger
(To cool down NO)

→

Oxidation chamber
(Conversion of NO into NO_2)

→

Absorption Tower
(warm water trickless)
dilute NO_2 into HNO_3

$$4NO_2 + 2H_2O + O_2 \rightarrow 4HNO_3$$

↓

Nitric acid (50%)



Properties of Nitric Acid

A. Physical Properties of Nitric Acid:

Nitric acid (HNO_3) is a strong mineral acid with several notable physical properties.

1. **Appearance:** Pure nitric acid is colourless. However nitric acid (impure) obtained in laboratory is slightly yellow or in brown colour. (NO_2)

2. **Odour:** Nitric acid has a distinctive and pungent smell.

3. **Taste:** Sour in taste.



Properties of Nitric Acid

A. Physical Properties of Nitric Acid:

Nitric acid (HNO_3) is a strong mineral acid with several notable physical properties.

4. **Density:** The density of nitric acid is approximately 1.54 g/cm^3.

5. **Melting point:** Nitric acid melts at –42°C.

6. **Boiling point:** It boils at 86°C.



Properties of Nitric Acid

A. Physical Properties of Nitric Acid:

Nitric acid (HNO_3) is a strong mineral acid with several notable physical properties.

7. **Solubility:** It is highly soluble in water forming a solution that is a strong acid.

8. **Corrosiveness:** Nitric acid is corrosive in nature and can react with many materials including metals.

like gold and Silver

Properties of Nitric Acid

$HNO_3 \rightleftharpoons H_3O^+ + NO_3^-$ → Monobasic acid

(One Hydronium ion per acid Molecule)

B. Chemical properties of nitric acid:

Nitric acid (HNO_3) exhibits various chemical properties reflecting its nature as a strong and highly reactive acid. Chemical properties of nitric acid are as follows:

1. **Acidic Acidic nature:** Nitric acid is a strong mono basic mineral acid. It dissociate.completely in water to produce hydronium (H_3O^+) and nitrate ions (NO_3^-).

Indicator	Original colour	Colour changes in HNO_3
Blue litmus paper	Blue	Red (Acidic)
Phenolphthalein	Colour less	Colourless
Methyl orange	Orange	Pink (Acid) OR Red

Properties of Nitric Acid

B. Chemical properties of nitric acid:

Nitric acid (HNO_3) exhibits various chemical properties reflecting its nature as a strong and highly reactive acid. Chemical properties of nitric acid are as follows:

2. **Thermal stability:** Pure nitric acid is unstable and decomposes even at normal temperature in presence of sunlight or on heating.

$$4HNO_3 \xrightarrow[\text{Sunlight}]{\text{heat}} 4NO_2 + 2H_2O + O_2$$

Properties of Nitric Acid

B. Chemical properties of nitric acid:

Nitric acid (HNO_3) exhibits various chemical properties reflecting its nature as a strong and highly reactive acid. Chemical properties of nitric acid are as follows:

3. Reaction with base: Nitric acid reacts with bases to form water and the corresponding nitrates salts. This reaction is called neutralisation reaction.

Metal Oxide ↓ Basic in Nature

Base + Acid (dilute) → Nitrate salt + water

$K_2O + 2HNO_3 \rightarrow 2KNO_3 + H_2O$

$Na_2O + 2HNO_3 \rightarrow 2NaNO_3 + H_2O$

$ZnO + 2HNO_3 \rightarrow Zn(NO_3)_2 + H_2O$

$CuO + 2HNO_3 \rightarrow Cu(NO_3)_2 + H_2O$

gmlp



Properties of Nitric Acid

B. Chemical properties of nitric acid:

Nitric acid (HNO_3) exhibits various chemical properties reflecting its nature as a strong and highly reactive acid. Chemical properties of nitric acid are as follows:

4. Reaction with carbonates and bicarbonates: Niric acid reacts with carbonates and bicarbonates, producing carbon dioxide, water and corresponding nitrate salt.

Metal carbonate/Bicarbonate + Acid Nitrate salt + Water + Carbon dioxide

$$Na_2CO_3 + 2HNO_3 \rightarrow 2NaNO_3 + H_2O + CO_2 \uparrow$$
$$MgCO_3 + 2HNO_3 \rightarrow Mg(NO_3)_2 + H_2O + CO_2 \uparrow$$

Salt + H_2O + CO_2 ↑

Metal carbonates are slightly Basic



Properties of Nitric Acid

B. Chemical properties of nitric acid:

Nitric acid (HNO_3) exhibits various chemical properties reflecting its nature as a strong and highly reactive acid. Chemical properties of nitric acid are as follows:

5. Reaction with sulphides and bisulphites: Nitric acid reacts with sulphite and bisulphites producing sulphur dioxide, water and corresponding nitrate salt.

Key concept: → Reactant

Product

Metal sulphite/Bisulphite + Acid → Nitrate salt + Water + Sulphur dioxide gas

$Na_2SO_3 + 2HNO_3 \rightarrow 2NaNO_3 + H_2O + SO_2 \uparrow$

$K_2SO_3 + 2HNO_3 \rightarrow 2KNO_3 + H_2O + SO_2 \uparrow$

$NaHSO_3 + HNO_3 \rightarrow NaNO_3 + H_2O + SO_2 \uparrow$

Properties of Nitric Acid

B. Chemical properties of nitric acid:

Nitric acid (HNO_3) exhibits various chemical properties reflecting its nature as a strong and highly reactive acid. Chemical properties of nitric acid are as follows:

6. .Oxidation of metals and non-metals: Nitric acid is a powerful oxidising agent and its oxidising properties are due to nascent properties and due to nascent oxygen. Depending upon the concentration and temperature, nitric acid undergoes to form NO, NO_2 and N_2O.

$$2HNO_3(conc) \xrightarrow{heat} 2NO_2 + H_2O + [O]$$

Decomposition of dilute nitric acid

$$2HNO_3(conc) \xrightarrow{heat} 2NO + H_2O + 3[O]$$

Properties of Nitric Acid

6. Oxidation of metals and non-metals:

i) Action on Metals:

a. With cold and dilute nitric acid:

Metal + Acid → Metals nitrate + Water + Nitric oxide

$3Zn + 8HNO_3 \rightarrow 3Zn(NO_3)_2 + 4H_2O + 2NO$

$3Zn + 8HNO_3 \rightarrow 3Cu(NO_3)_2 + 4H_2O + 2NO$

dil. HNO_3 → No (Nitric oxide)

Metal + Nitric acid (dil.) → Metal nitrate + H_2O + No (Nitric oxide)



Properties of Nitric Acid

6. Oxidation of metals and non-metals:

i) Action on Metals:

b. With concentrated nitric acid (or hot nitric acid)

$$Zn + 4HNO_3 \rightarrow Zn(NO_3)_2 + 4H_2O + 2NO_2$$

$$Cu + 4HNO_3 \rightarrow Cu(NO_3)_2 + 4H_2O + 2NO_2$$

$$Fe + 6HNO_3 \rightarrow Fe(NO_3)_3 + 3H_2O + 3NO_2$$

HNO_3 (conc.) $\longrightarrow NO_2$

Properties of Nitric Acid

6. Oxidation of metals and non-metals:

i) Action on Metals:

c. With very dilute nitric acid:

Magnesium and manganese react with very dilute nitric acid to form nitrate salt and hydrogen. gas

$$Mg + \underset{\text{(very dilute)}}{2HNO_3} \rightarrow Mg(NO_3)_2 + H_2$$

$$Mn + \underset{\text{(very dilute)}}{2HNO_3} \rightarrow Mn(NO)_2 + H_2$$

gmb

Properties of Nitric Acid

6. Oxidation of metals and non-metals:

ii) Action on non-metals:

Non-metal + conc. acid →

Oxidised product (acid) + Water + Nitrogen dioxide.

$C + 4HNO_3 \rightarrow CO_2 + 2H_2O + 4NO_2$

$P + 5HNO_3 \rightarrow H_3PO_4 + 2H_2O. + 5NO_2.$

$S + 6HNO_3 \rightarrow H_2SO_4 + 2H_2O + 6NO_2$

(Conc.)

Oxidised Products :- CO_2, H_3PO_4, H_2SO_4

HNO_3 (Conc.)

gen. Rxn : Non Metal + HNO_3 (Conc.) ⟶ oxidised product + H_2O + NO_2

Properties of Nitric Acid

7. Reaction with hydrochloric acid:

When concentrated nitric acid reacts with hydrochloric acid (in the ratio 1:3), it forms a mixture known as aqua regia. (1:3)

$$HNO_3 + HCl \rightarrow NOCl + 2H_2O + 2[Cl]$$

(Nitrosyl chloride)

Imp

$$Au + 3[Cl] \rightarrow AuCl_3$$

Gold (III) chloride

$$Pt + 4[Cl] \rightarrow PtCl_4$$

Platinum (IV) chloride

Aqua Regia: Royal water

↓

(It can dissolve gold)

Note:

- Aqua regia is a mixture of conc. HNO_3 and conc. HCl in the ratio (1 : 3). It is called royal water.

- Nitric acid oxidises hydrochloric acid to chlorine gas → cause dissolution of gold and Pt

HNO_3 (1) : HCl (3) → Aqua Regia

1 : 3

HNO_3 — HCl

Effects of Heat on Nitrates

1. Decomposition of sodium and potassium nitrate (alkali metal nitrate)

$$2NaNO_3 \xrightarrow{\Delta} 2NaNO_2 + O_2$$

$$2KNO_3 \xrightarrow{\Delta} 2KNO_2 + O_2$$

Thermal decomposition of Metal nitrate

↓

Metal nitrite + O_2 (g)

Effects of Heat on Nitrates

2. Decomposition of calcium nitrate, copper nitrate, lead nitrate and zinc. nitrate.

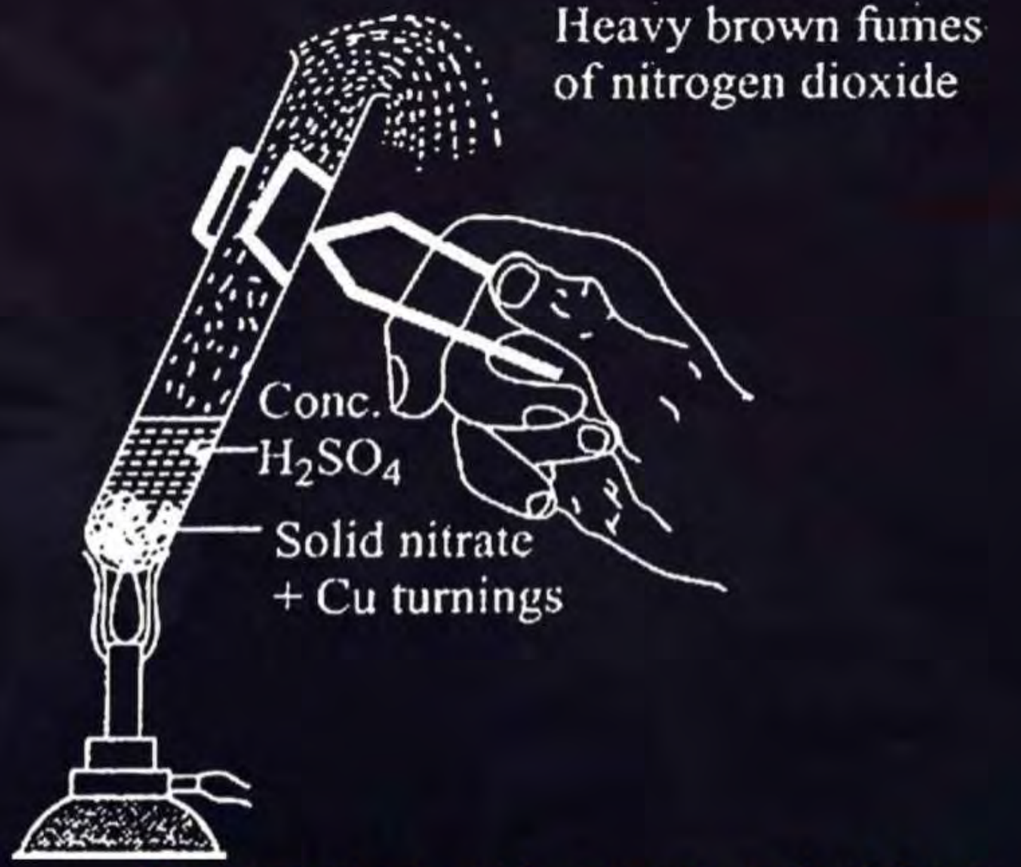
$$2Ca(NO_3)_2 \xrightarrow{\text{Heat}} 2CaO + 4NO_2 + O_2$$

crystalline (colourless) → (white) CaO; raddish (brown gas) NO_2

$$2Cu(NO_3)_2 \xrightarrow{\text{Heat}} 2CuO + 4NO_2 + O_2$$

crystalline (colourless) → $Cu(NO_3)_2$

(black) CuO; (Raddish Brown fumes)



Heating a nitrate with concentrated sulphuric acid and Cu turning (heavy, brown fumes of nitrogen dioxide are evolved)

Effects of Heat on Nitrates

3. **Decomposition of ammonium nitrate: When ammonium nitrate (NH_4NO_3) is heated it undergoes thermal decomposition.**

$$\underset{\text{crystalline (colourless)}}{NH_4NO_3} \xrightarrow{\text{Heat}} \underset{\text{nitrous oxide}}{N_2O_{(g)}} + \underset{\text{(vapour)}}{2H_2O}$$

(laughing gas)

N_2O

Test for Nitric Acid and Nitrates

1. By heating

i) On heating concentrated nitric acid gives brown fumes.

$$4HNO_3 \xrightarrow{\text{Heat}} 4NO_2 + 2H_2O + O_2$$

(Brown fumes)

ii) All nitrates (other than sodium and potassium and ammonium) on heating produces reddish brown fumes of nitrogen dioxide (NO_2).

$$2Cu(NO_3)_2 \xrightarrow{\text{Heat}} 2CuO + 4NO_2 + O_2$$

(raddish brown fumes)

Test for Nitric Acid and Nitrates

2. By heating with copper:

i) When nitric acid or nitrate solution is heated with copper metal, dense reddish brown fumes of nitrogen dioxide are formed. (NO_2).

$$Cu + 4HNO_3 \xrightarrow{\text{Heat}} Cu(NO_3)_2 + \underset{\substack{\text{raddish}\\\text{brown}\\\text{fumes}}}{4NO_2} + 2H_2O$$

Test for Nitric Acid and Nitrates

3. Potassium iodide test:

When we pass nitric acid vapours to a filter paper strip soaked in KI solution it turms brown.

$$4HNO_3 + 2KI \rightarrow 2KNO_3 + 2H_2O + \boxed{2NO_2} + I_2$$

Test for Nitric Acid and Nitrates

3. Brown ring test:

imp.

The brown ring test is a chemical test used to detect the presence of nitrates in a solution.

Procedure:

i) Take 5 ml aqueous solution of a nitrate or nitric acid.

ii) Add a few drops of freshly prepared iron (II) sulphate ($FeSO_4$) solution in it.

Nitric Acid (aq) + Freshly Prepa. $FeSO_4$

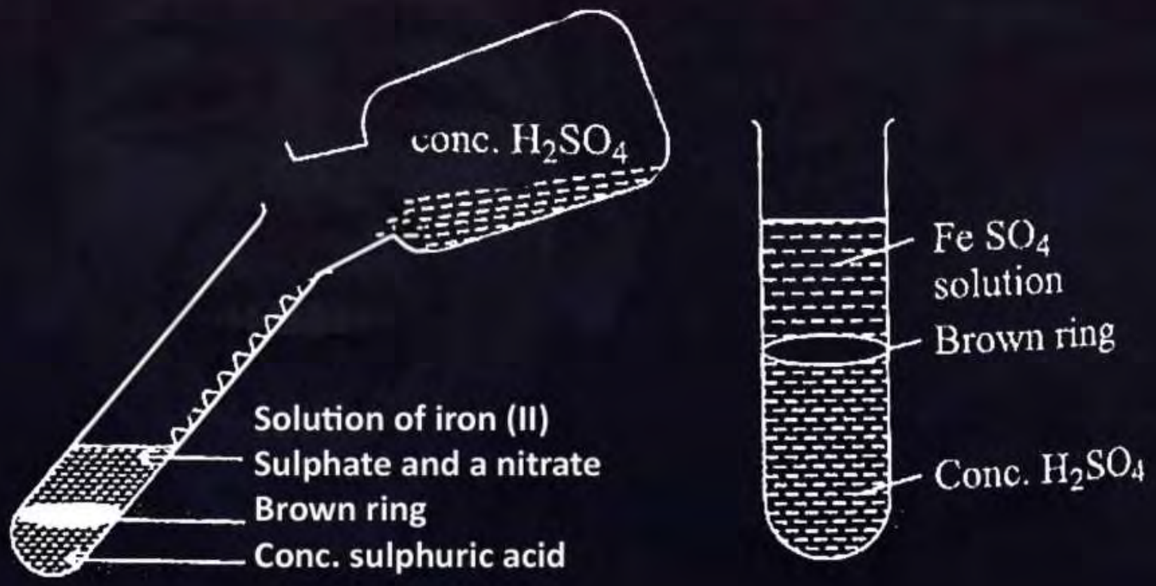
Test for Nitric Acid and Nitrates

3. Brown ring test:

The brown ring test is a chemical test used to detect the presence of nitrates in a solution.

Procedure:

iii) Now add conc. sulphuric acid (H_2SO_4) down the side of test-tube without mixing two layers.



Test for Nitric Acid and Nitrates

Procedure:

iii) Now add conc. sulphuric acid (H_2SO_4) down the side of test-tube without mixing two layers.

Observation: A brown ring forms at the junction of two layers.

Imp.

Chemical reaction:

$$6FeSO_4 + 3H_2SO_4 + 2HNO_3 \rightarrow 3Fe_2(SO_4)_3 + 4H_2O + 2NO$$

Ferrous (II) sulphate

Iron (III) sulphate (raddish brown)

$$FeSO_4 + NO \rightarrow FeSO_4.NO$$

Nitroso ferrous sulphate (a brown ring)

Imp.

Test for Nitric Acid and Nitrates

Procedure:

i) Do not disturb the test tube otherwise brown ring decomposes. It is due to the heat evolved when conc. H_2SO_4 and water are mixed together.

★ Imp

ii) Use freshly prepared ferrous sulphate solution because on keeping it to the atmosphere, ferrous sulphate is oxidized to ferric sulphate which will not form brown ring.



Uses of Nitric Acid

i) **Chemical manufacturing:** It is a key ingredient in the production of various chemicals, fertilizers and explosives.

ii) **Fertilizer production:** It is used in making ammonium nitrate, calcium nitrate, nitro-chalk [NH_4NO_3 + $CaCO_3$] and basic calcium nitrate $CaO.Ca(NO_3)_2$.

iii) **Explosive:** Nitric acid is a crucial component in the production of explosives such as trinitro toluene (IN.T.), nitroglycerine and picric acid.



Uses of Nitric Acid

iv) **Dyes and pigments:** Nitric acid is involved in the production of dyes, pigments and some types of ink.

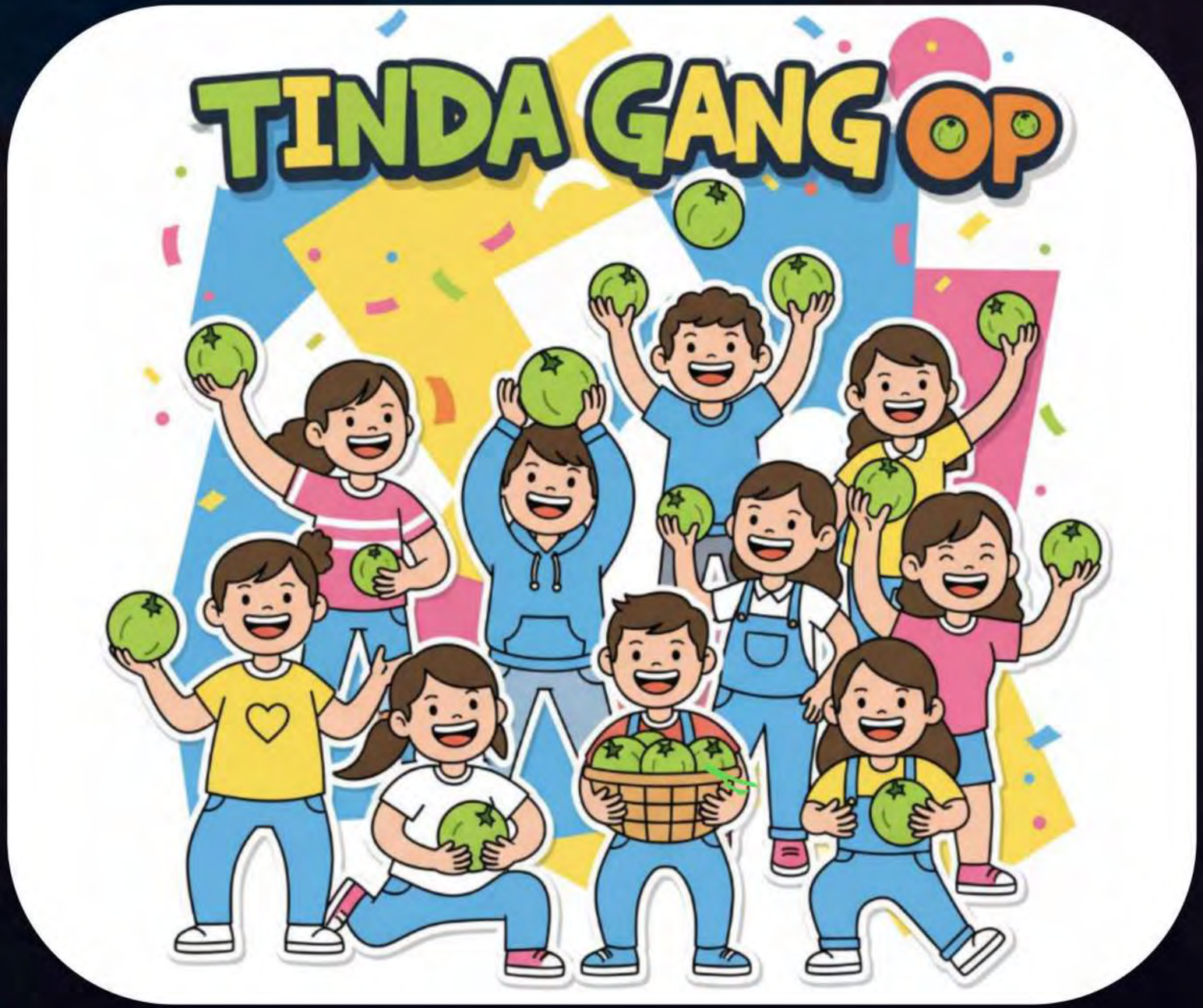
v) **Photography:** Employed in the production of some photographic chemicals.

Question

What is aqua fortis?




BHARTI MAAM
JOIN MY OFFICIAL TELEGRAM CHANNEL
Physics Wallah
@CHEMISTRYBYBHARTIMAAM


TINDA GANG OP


Thank You